

Calculate relative frequencies from given or collected data to estimate probabilities of events involving “and”, inclusive “or” and exclusive “or”

Learning intention

- calculate relative frequencies from given or collected data to estimate probabilities of events involving “and”, inclusive “or” and exclusive “or”.

Teach and model

- using Venn diagrams or two-way tables to estimate frequencies of events involving “and”, “or” questions.
- designing, testing and refining an algorithm used to determine relative frequencies from a generated data set, to estimate different probabilities.
- understanding that relative frequencies from large data sets or long-run experiments can provide reliable measures of probability and can be used to make predictions of decisions.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.